

Next Prime

Time limit: 1.00 s

Memory limit: 512 MB

Given a positive integer n , find the next prime number after it.

Input

The first line has an integer t : the number of tests.

After that, each line has a positive integer n .

Output

For each test, print the next prime after n .

Constraints

- $1 \leq t \leq 20$
- $1 \leq n \leq 10^{12}$

Example

Input	Output
5	2
1	3
2	5
3	43
42	1361
1337	